

GRPO

What is Reference Tuning?

What is RL?

What is a PPO?

What is RLHF?

What is GRPO?

GRPO (Group Relative Policy Optimization) is a reinforcement learning method used in LLM alignment that optimizes model behavior based on relative comparisons within a group of responses, instead of relying on absolute reward scores.

GRPO learns by comparing multiple responses and improving based on relative ranking instead of absolute reward values.

Key Idea

- PPO → optimizes using absolute reward values
- GRPO → optimizes using relative ranking within a group

Instead of asking:

"How good is this answer?"

GRPO asks:

"Which answer is best among this group?"

Then GRPO (2024–2025, used by DeepSeek)

👉 Evolution of PPO + DPO ideas

Group-based comparison + RL optimization

👉 Key innovation

No reward model + multiple samples per prompt

Why GRPO?

- More stable than PPO
- Less dependent on precise reward modeling
- Better suited for LLM alignment scenarios
- Works well when:
 - evaluation rule exists
 - multiple samples possible

Positioning

- PPO → traditional RL method (still used, especially in RLHF)
- GRPO → newer alternative focusing on relative comparison
- DPO / ORPO → often preferred today (no RL loop, simpler training)

GRPO is a promising alternative to PPO in certain alignment setups

How GRPO Works

1. Take a prompt
2. Generate multiple responses (e.g., 4–8)
3. Compare or rank the responses
4. Compute relative advantage within the group
5. Update the model based on relative performance

Core Intuition

- PPO → “Assign a score to each answer”
- GRPO → “Compare answers and learn which is better”

How GRPO Works (Step-by-Step)

Step 1: Given a prompt

User: Explain AI

Step 2: Generate multiple responses

A: AI is simulation of intelligence

B: AI is random stuff

C: AI is machines learning patterns

Step 3: Compare within group

Instead of reward model:

Rank them:

$C > A > B$

Step 4: Compute relative advantage

GRPO computes:

“How much better is A compared to others?”

Based on group ranking

Step 5: Update policy

Increase probability of better responses

Decrease probability of worse ones

Similar to PPO

But no reward model

Intuition (Super Simple)

PPO:

"This answer = score 8/10"

GRPO:

"This answer is better than others"

Real Example

Prompt:

Write a polite email

Generated:

- A → rude
- B → okay
- C → very polite

GRPO learns:

$C > B > A$

Update:

- Increase C
- Decrease A

Why DeepSeek Used GRPO

Problem DeepSeek faced

They were training huge LLMs (DeepSeek-R1)

PPO issues:

- Reward model training cost
- Instability
- Slow

Problem with PPO (RLHF)

- Needs Reward Model
- Expensive training
- Complex pipeline

DeepSeek Solution = GRPO

Instead of:

Generate → Reward Model → PPO

They did:

Generate multiple responses → Compare → Update

🔥 CRITICAL FIX (DeepSeek part)

❌ Your statement:

- "No reward model + self learning only"

✅ Correct:

- DeepSeek used rule-based + task-specific rewards
- Not purely "model learning from itself blindly"

👉 Example:

- math problems → correctness check
- reasoning → format / logic check

❌ Your statement:

- "GRPO = no reward model"

✅ Correct version:

- GRPO can work without a trained reward model
- BUT still needs some scoring signal (heuristic / rule / proxy reward)

👉 Matlab:

- Reward model ❌
- Reward signal ✅

MOST IMPORTANT CORRECTION (Conceptual)

❌ Wrong understanding:

- "GRPO = no reward"

✅ Correct:

- GRPO still needs signal to compare outputs
- It just avoids training a separate reward model

Why GRPO is better (for them)

1. No Reward Model

- Removes entire pipeline

2. Self-improving loop

Model generates multiple answers → learns from its own outputs

3. More stable

- Uses relative advantage
- Not absolute noisy reward

4. Cheaper & faster

No reward model training

Less infrastructure

⚠️ WHERE YOU WERE SLIGHTLY WRONG / INCOMPLETE

❌ 1. "No reward model" ko galat samjha

👉 Tu bola:

GRPO = no reward model

👉 Correct:

GRPO = no trained reward model

BUT still needs reward signal (score)

✓ Paper me:

DeepSeek used rule-based reward (correct/incorrect)

❌ 2. Value model removal ka exact reason missing tha

👉 Tu bola:

GRPO simpler hai

👉 Paper actually bolta hai:

Value model remove karke

group mean se advantage calculate hota hai

👉 Ye core innovation hai

❌ 3. Advantage formula intuition missing

👉 Tu bola:

"relative advantage"

👉 Paper me:

Proper normalization hota hai:

👉 $(\text{reward} - \text{mean}) / \text{std}$

✓ Ye hi "group relative" ka actual meaning hai

❌ 4. KL penalty placement nuance missing

👉 Tu bola:

similar to PPO

👉 Correct:

PPO → KL inside advantage

GRPO → KL separate term

👉 Small but important research detail

1. Setup and notation

Maan le:

- q = prompt / question
- π_θ = current policy model (LLM with parameters θ)
- $\pi_\theta(\text{"old"})$ = old policy
- $\pi_{\text{"ref"}}$ = reference model
- G = number of sampled outputs for one prompt
- o_i = i-th sampled output
- r_i = reward for output o_i

For one prompt q, GRPO samples:

$o_1, o_2, \dots, o_G \sim \pi_\theta(\text{"old"}) (-lq)$

Matlab ek prompt ke liye ek answer nahi, multiple answers generate karte hain. Ye GRPO ka core hai.

2. Reward collection

Har sampled output ka reward nikalte hain:

$r_i = R(q \oplus o_i)$

Yahan R ho sakta hai:

- trained reward model
- rule-based reward
- correctness checker
- format checker

DeepSeek-style setup me trained reward model zaroori nahi; reward rules se bhi aa sakta hai.

3. Group mean and group standard deviation

GRPO ka main innovation ye hai ki ye value model use nahi karta.

Uski jagah ye group ke rewards ka mean aur standard deviation nikalta hai:

$\mu_r = 1/G \sum_{i=1}^G r_i$

$\sigma_r = \sqrt{1/G \sum_{i=1}^G (r_i - \mu_r)^2}$

Yahi wo "group statistics" hain jo value model ko replace karte hain. Paper explanation bhi yehi bolta hai: rewards ko average reward aur standard deviation se normalize kiya jata hai, so group khud baseline ban jata hai.

4. Relative advantage in GRPO

Ab har output ka relative advantage nikalte hain:

$A_i = (r_i - \mu_r) / \sigma_r$

Kabhi kabhi numerical stability ke liye:

$A_i = (r_i - \mu_r) / (\sigma_r + \epsilon)$

Yahan:

- agar $r_i > \mu_r$, then $A_i > 0$
- agar $r_i < \mu_r$, then $A_i < 0$

Matlab:

- good answer $\rightarrow$ positive advantage
- bad answer $\rightarrow$ negative advantage

Yehi "group relative" ka real meaning hai. GRPO group ke andar compare karta hai, absolute score par depend nahi hota.

5. Policy ratio

Ab PPO ki tarah policy ratio define hota hai:

$\rho_i(i, t) (\theta) = (\pi_\theta(o_i(i, t) \oplus lq_o(i, < t))) / (\pi_\theta(\text{"old"})(o_i(i, t) \oplus lq_o(i, < t)))$

Yahan:

- $o_i(i, t)$ = output o_i ka t-th token
- numerator = new policy ki probability
- denominator = old policy ki probability

Agar ratio > 1:

- new model us token ko zyada prefer kar raha hai

Agar ratio < 1:

- new model us token ko kam prefer kar raha hai

Paper explanation me bhi isi ratio ko policy ratio bola gaya hai.

6. Clipped surrogate term

PPO ki tarah GRPO bhi clipping use karta hai, taaki update bahut large na ho:

$\min_i \frac{1}{G} (\rho_i(i, t) (\theta) A_i + \text{clip}(\rho_i(i, t) (\theta) \otimes, "1 - \epsilon", "1 + \epsilon") A_i)$

Yahan ϵ clipping hyperparameter hai, usually small value.

Intuition:

- agar answer achha hai, model uski probability badhaye
- lekin ekdum se bahut zyada nahi
- clipping training ko stable rakhti hai

Paper explanation me clipping ko stable incremental improvement ke liye bataya gaya hai.

7. KL penalty with reference model

GRPO me KL penalty still hoti hai, lekin PPO ke advantage ke andar nahi hoti; alag term ki tarah aati hai. Paper explanation explicitly ye point karta hai.

Ek common token-level KL form:

$D_{\text{"KL"}}(\pi_\theta \parallel \pi_{\text{"ref"}})$

Practical training me isko beta se weight kiya jata hai:

$\beta D_{\text{"KL"}}(\pi_\theta \parallel \pi_{\text{"ref"}})$

Yahan:

- $\pi_{\text{"ref"}}$ = reference model
- β = KL penalty weight

Purpose:

- model ko reference se bahut dur drift hone se rokna

8. Full GRPO objective

Ab one prompt aur one group ke liye intuitive GRPO objective likh sakte hain:

$L^{\text{GRPO}}(\theta) = -1/G \sum_{i=1}^G \frac{1}{(l_{o_i})} \sum_{t=1}^T (l_{o_i}) \min_i \frac{1}{G} (\rho_i(i, t) (\theta) A_i + \text{clip}(\rho_i(i, t) (\theta) \otimes, 1 - \epsilon + \epsilon, 1 + \epsilon) A_i) + \beta D_{\text{"KL"}}(\pi_\theta \parallel \pi_{\text{"ref"}})$

Isko samajhne ka best way:

- outer sum over i: group ke multiple answers
- inner sum over t: token-level learning signal
- clipped term: stable PPO-style update
- KL term: safe update

Paper explanation ke mutabik GRPO multiple outputs ke objectives average karta hai, token-level signal propagate karta hai, aur KL ko separate handle karta hai.

How to grade answers

